INDIAN STATISTICAL INSTITUTE
Mid-Semester Examination: 2025-26 (First Semester)

M. MATH. II YEAR
Commutative Algebra I

Date : 9.9.25 | Maximum Marks : 40 | Duration : $2\frac{1}{2}$ Hours

Throughout R denotes a commutative ring with unity.

ANSWER ANY FIVE QUESTIONS.
Each question carries 9 marks.

1. (i) Compute $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Q}$ and $\mathbb{C} \otimes_{\mathbb{R}} \mathbb{C}$.
(ii) Give an example to show that the natural map $(\prod_n M_n) \otimes_R N \to \prod_n (M_n \otimes_R N)$ is not an isomorphism.

2. State Nagata's criterion for an integral domain to be UFD. Prove that the ring $\mathbb{C}[X,Y,Z]/(X^2+Y^3+Z^7)$ is a UFD.

3. Let $A = \mathbb{R}[X,Y]/(X^2+Y^2-1)$. Let x and y denote the images of X and Y in A and $P = (x, y-1)$. Prove that P is a projective A-module.

4. Prove that any finitely generated flat module over a local ring is free. Give an example to show that the finite generation hypothesis is necessary.

5. Prove that any finitely generated subring of $\mathbb{Q}$ is of the form $\mathbb{Z}[1/a]$ for some $a \in \mathbb{Z} \setminus \{0\}$.

6. Let $B = \mathbb{C}[X,Y,Z]/(XY-Z^2)$ and $A = \mathbb{C}[T^2, T^3]$. Prove that there exists a surjective $\mathbb{C}$-algebra map ϕ from B to A. Compute $\ker \phi$.

7. Prove that $\mathbb{C}[X,Y]/(X^2+Y^2-1)$ is a PID.

8. Prove that an integral domain R is a UFD if and only if every prime ideal of R contains a prime element. Deduce that if R is a PID, then $R[[X]]$ is a UFD using Kaplansky's result.

9. (i) Prove that any surjective R-linear map $f : M \to M$ of a finitely generated R-module M is an automorphism.
(ii) Give an example of a finitely presented R-module M and an injective R-linear map $\phi : M \to M$ which is NOT surjective.